The Zhou–Li rate for iterative spectral likelihood decoding

Research note for the spectral-lrt project

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Abstract

We analyze the full Bernoulli spectral-profile decoder, rather than a raw-adjacency refinement. An objective-certified fit already has E-step error $O((np)^{-1})$ on the same exponentially reliable graph event. Its mandatory M-step permits a constant-scale score comparison. The central additional ingredient is an arrowhead expansion conditioning on the graph with one vertex removed. It converts the full-graph spectral block sums into linear functions of independent incident edges with a bounded remainder. A finite-branch treatment of clipping and a tilted local concentration bound recover the factor $(np)^{-1/2}$. Consequently the certified algorithm attains a constant-factor oracle rate without an additional change to its score or updates. For a merely weak responsibility initializer without an objective certificate, a two-M-step local refinement is sufficient. The exponential-confidence spectral specialization and the replacement objective certificate are verified explicitly. The new arguments have undergone independent internal mathematical checks.

1 Model, algorithm, and exact target

The partition $z \in [K]^n$ is a fixed parameter. The graph is undirected, has no loops, and its independent upper-triangular edges satisfy

$$A_{ij} \sim \text{Bernoulli}(P_{z_i z_j}), \qquad p = \max_{a,b} P_{ab}.$$

The known K is fixed. For fixed positive π_0, κ, η, c assume

$$n_a \geq \pi_0 n, \quad \kappa p \leq P_{ab} \leq p \leq 1 - \eta, \quad \sigma_{\min}(P) \geq \kappa p, \quad np \geq c \log n. \tag{1}$$

No convergence of P/p or the proportions is required. Write $Z_{ia} = \mathbf{1}\{z_i = a\}$, $M_0 = ZPZ^\top$, and $\Pi = Z(Z^\top Z)^{-1}Z^\top$. The actual mean is $M = M_0 - \text{diag}(P_{z_i z_i})$.

Let $F = A_K = U\Lambda U^\top$ retain the K eigenpairs of largest absolute eigenvalues. Put

$$s = \max(1, \|\Lambda\|_{\text{op}}), \qquad \epsilon = c_0 s/n, \qquad q_{ij} = \text{clip}(F_{ij}, \epsilon, 1 - \epsilon),$$

where $0 < c_0 < \min(\kappa, \eta, 1)/8$. The free-profile score is

$$\ell_i(\mu) = \sum_j \{q_{ij} \log \mu_j + (1 - q_{ij}) \log(1 - \mu_j)\}.$$

The M-step and E-step are

$$\mu_{aj} = \frac{\sum_l R_{la} q_{lj}}{\sum_l R_{la}}, \quad \pi_a = \frac{1}{n} \sum_l R_{la}, \qquad R_{ia}^{\text{new}} = \frac{\pi_a e^{\ell_i(\mu_a)}}{\sum_b \pi_b e^{\ell_i(\mu_b)}}.$$

All these operations use only the retained eigenpairs. In proofs the true labels are used solely for comparison.

For $z_i = a$ and $b \neq a$, set

$$w_c^{ab} = \log \frac{P_{ac}(1 - P_{bc})}{P_{bc}(1 - P_{ac})}, \qquad b_c^{ab} = \log \frac{1 - P_{ac}}{1 - P_{bc}}.$$

The oracle incident-edge LLR is

$$L_i^{ab} = \sum_{j \neq i} \{A_{ij} w_{z_j}^{ab} + b_{z_j}^{ab}\}.$$

Let $I_{i,ab} = -\log \min_{0 \leq \alpha \leq 1} \mathbb{E}_a e^{-\alpha L_i^{ab}}$ and $I_n = \min_{i, a \neq b} I_{i,ab}$. Under (1), $I_{i,ab} \asymp np$, $\alpha_{i,ab}^* \in [\alpha_0, 1 - \alpha_0]$, and the tilted variance is $\asymp np$. Thus the Zhou–Li binary oracle scale is $e^{-I_{i,ab}}/\sqrt{np}$, uniformly up to constants.

Lemma 1 (Uniform oracle constants)**.** *The preceding information, maximizing-parameter, and variance bounds hold uniformly over* (1)*. The equal-prior binary oracle error is between constant multiples of* $e^{-I_{i,ab}}/\sqrt{np}$*.*

Proof. The odds weights are uniformly bounded and, for $u, v \in [\kappa p, p]$ with $p \leq 1 - \eta$, their absolute difference in logits is between $|u - v|/p$ and $|u - v|/(\kappa \eta p)$. Distinct rows of P are separated in Euclidean norm by at least $\sqrt{2}\sigma_{\min}(P)$. Thus $\sum_c (n_c - \mathbf{1}\{c = a\})(w_c^{ab})^2 \asymp n$. At any tilt $0 \leq \alpha \leq 1$, its Bernoulli probabilities lie between P_{ac} and P_{bc}. Their variances are therefore between constant multiples of p, so the second derivative of $\psi(\alpha) = \log \mathbb{E}_a e^{-\alpha L_i^{ab}}$ is between $c_1 np$ and $C_1 np$. Since $\psi(0) = \psi(1) = 0$, integrating this curvature gives information $\asymp np$ and an interior optimizer bounded away from both endpoints. Bounded log-odds weights imply that the sum of tilted third absolute centered moments is at most a constant times the tilted variance. The rate-optimal Chernoff bounds of Zhou–Li then give the two-sided oracle rate. □

2 Explicit spectral input and refinement theorems

We use the following input already underlying the current manuscript. It is stated separately to make the dependency of this extension visible.

Proposition 2 (High-confidence spectral input)**.** *For each fixed* $H > 0$*, the graph and each principal subgraph obtained by deleting one vertex admit good events with failure at most* $C_H e^{-Hnp}$*. On these events the centered operator norm is at most* $C_H\sqrt{np}$*, the retained signal eigenvalues have magnitudes between* $c_H np$ *and* $C_H np$*, their orthonormal eigenvector matrices have maximum row norm* $C_H/\sqrt{n}$*, and their projectors differ from the true membership projectors by at most* $C_H/\sqrt{np}$ *in operator norm. The full graph also has maximum degree at most* $C_H np$*,* $\epsilon \leq 2c_0 p$*, and all* q_{ij} *lie in a common box* $[c_H p, \min(C_H p, 1 - b_H)]$ *with* $b_H > 0$*.*

Proof. The actual mean M leaves the membership space invariant. In its orthonormal membership basis its restriction is

$$(Z^\top Z)^{1/2} P (Z^\top Z)^{1/2} - \operatorname{diag}(P_{11}, \ldots, P_{KK});$$

on the orthogonal complement its eigenvalues are the numbers $-P_{aa}$. Consequently the K signal eigenvalues have magnitudes between $c_1 np$ and $C_1 np$, their projector is exactly Π, and their population row norms are $O(n^{-1/2})$.

The bounded-entry matrix norm inequality of Bandeira–van Handel (Corollary 3.12 together with Remark 3.13 for centered, not necessarily symmetrically distributed entries) gives, after absorbing $\sqrt{\log n}$ into $\sqrt{np}$,

$$\mathbb{P}\{\|A - M\|_{\mathrm{op}} > C\sqrt{np} + t\} \le n e^{-t^2/C}.$$

Taking $t = C_H\sqrt{np}$ provides any prescribed fixed confidence e^{-Hnp}, since $\log n \le c^{-1}np$. A degree Chernoff bound with a sufficiently large fixed multiplier gives the maximum-degree assertion at the same confidence.

For completeness, we verify the confidence parameter in the entrywise eigenspace theorem of Abbe–Fan–Wang–Zhong, rather than replacing an exponential confidence by a polynomial one. In their Theorem 2.1 use

$$\gamma = \frac{C_H}{\sqrt{np}}, \qquad \varphi_H(x) = \frac{C_H}{1 \vee \log(1/x)}\ (x > 0), \qquad \varphi_H(0) = 0, \quad \delta_0, \delta_1 \le e^{-Hnp}.$$

Their Bernoulli row inequality (Lemma 7) holds for every fixed nonnegative row-tail parameter. Choose that parameter sufficiently larger than $H + c^{-1}$. It gives per-row failure at most δ_1/n. Applied to each of the fixed number of columns, the monotonicity of φ_H and of $\varphi_H(x)/x$ verifies their matrix row-concentration assumption. Factors depending on K and the signal gap are included in C_H. Row/column independence follows from the independent upper-triangular edges. Their mean-row condition follows from $\|M\|_{2\to\infty} \le np/\sqrt{n}$ and $np \le n$. Their smallness condition holds eventually because $\gamma \to 0$ and $\varphi_H(\gamma) \to 0$.

Apply the theorem separately to the positive and negative signal groups, each separated by a gap of order np from its complement. Its row-norm bound gives $\|U\|_{2\to\infty} \le C_H/\sqrt{n}$. The theorem does not require M to have exactly rank K, so the no-loop diagonal correction is retained. The operator norm event and the signal gap give the asserted projector bound by the usual subspace perturbation inequality. A fixed number of union bounds is absorbed by increasing the confidence parameter in the preceding steps.

The Perron eigenvalue of M lies between its minimum and maximum row sums, which are in $[\kappa(n-1)p, np]$. The norm event gives $\kappa np/4 \le s \le 2np$ eventually. Thus $\epsilon \asymp p$, every true probability is at distance at least $c'p$ from both clipping endpoints, and

$$|F_{ij}| \le \|U_i\|_2 \|\Lambda\|_{\mathrm{op}} \|U_j\|_2 \le C_H p.$$

This proves the common box: if $C_H p \le 1/2$ it is bounded away from one directly; otherwise $\epsilon \ge c_H p \ge c_H/(2C_H)$ gives a fixed upper gap. Removing one vertex preserves all model bounds with adjusted fixed constants, and the same argument proves the vertex-deleted assertions uniformly in i. In a per-node risk calculation one only needs the event for that particular deleted vertex. □

Theorem 3 (Refined rate from weak responsibilities)**.** *Assume* (1) *and Proposition 2. Suppose a responsibility matrix $R^{(0)}$, using only the retained spectrum, has aligned average error bounded by a deterministic sequence tending uniformly to zero on a graph event whose failure is at most $C_H e^{-Hnp}$, for a sufficiently large fixed H. Perform*

$$R^{(0)} \xrightarrow{M_1} (\mu^{(1)}, \pi^{(1)}) \xrightarrow{E_1} R^{(1)} \xrightarrow{M_2} (\mu^{(2)}, \pi^{(2)}),$$

and classify by $\ell_i(\mu_a^{(2)}) + \log \pi_a^{(2)}$. Then

$$\mathbb{E}\,\mathrm{Mis}(\widehat{z}, z) \le C\frac{e^{-I_n}}{\sqrt{np}}. \tag{2}$$

The same bound holds after additional finite exact E/M updates. Constants may depend on the fixed model bounds and on K, not on n.

For a merely weak responsibility initializer, this local statement uses two M-steps. The complete growing-and-replacement algorithm has a stronger objective certificate: its endpoint E-step is already accurate enough, so the existing single mandatory final M-step suffices. The next section states and proves this improvement; the remaining sections supply the refined tail analysis common to both statements.

3 One update supplies the required initialization accuracy

Lemma 4 (Frobenius error and responsibility bootstrap)**.** *On the full-graph event of Proposition 2,*

$$\|F - M_0\|_{\mathrm{F}}^2 \le Cnp, \qquad \|q - M_0\|_{\mathrm{F}}^2 \le Cnp.$$

If the fitted profiles have aligned ℓ_1 error $o(np)$ and $-\log \min_a \pi_a = o(np)$ (in particular, if the weights are bounded below), their next E-step satisfies

$$\frac{1}{n}\sum_{i,a} |R_{ia}^{\mathrm{new}} - Z_{ia}| \le \frac{C}{np}.$$

Proof. Best rank-K approximation in operator norm gives $\|A-F\|_{\mathrm{op}} \le \|A-M_0\|_{\mathrm{op}} \le C\sqrt{np}$. The matrix $F - M_0$ has rank at most $2K$, proving the Frobenius bound. True probabilities lie inside the clipping interval, so clipping cannot increase squared coordinate error.

The population gap $D(P_{a*}\|P_{b*})$ is at least $c_1 np$. On the common box, changing an observation or a parameter changes its score contrast by at most a constant times its ℓ_1 change. Thus a row with $\|q_i - P_{a*}\|_1 \le t_0 np$, for sufficiently small fixed t_0, has fitted correct-class margin at least $c_2 np$ for all large n. Its wrong responsibility is at most $(K-1)e^{-c_2 np}$. Every other row has squared ℓ_2 error at least $t_0^2 np^2$. The Frobenius bound limits their number to C/p, hence their fraction to $C/(np)$. □

An initial M-step from $o(1)$ aligned responsibilities has profile error $o(np)$, by the original mean-fitting argument. Therefore Lemma 4 gives $R^{(1)}$ with error $C/(np)$. This refined radius remains invariant: an M-step from this radius has profile error at most $C\sqrt{np} + C = o(np)$ in ℓ_1. More quantitatively, choose a fixed small responsibility radius so that the resulting parameter error costs at most one quarter of the population margin. Every E-step from that fixed radius has error at most $C_*/(np)$, where C_* is independent of the previous error and of the iteration number. Since $C_*/(np)$ eventually lies inside the fixed radius, induction gives a common bound for all subsequent updates.

Lemma 5 (An objective certificate already gives the sharp initialization radius)**.** *Suppose a K-profile fit is in the common box and its soft loss satisfies*

$$\mathcal{F} = -\sum_i \log \sum_a \pi_a e^{-D(q_i\|\mu_a)} \le \xi_n n^2 p, \qquad \xi_n \longrightarrow 0,$$

on the good graph event, with a deterministic uniform sequence ξ_n. Then its profiles admit a common matching to the true profiles with maximum ℓ_1 error $o(np)$, its weights satisfy $-\log \min_a \pi_a = o(np)$, and its own exact E-step has aligned error at most $C/(np)$.

Proof. Write $d = np$. Since $\mathcal{F} \geq \sum_i \min_a D(q_i \| \mu_a)$ and Bernoulli divergence on the box is at least $\|q_i - \mu_a\|_1^2/(Cd)$,

$$\sum_i \min_a \|q_i - \mu_a\|_1^2 \leq C\xi_n n d^2.$$

Also $\sum_i \|q_i - P_{z_i,*}\|_1^2 \leq n\|q - M_0\|_{\mathrm{F}}^2 \leq Cnd$. Choose $v_n = \xi_n^{1/4} + d^{-1/4} \to 0$. The fractions of rows farther than $v_n d$ from their true profile or from every fitted profile are at most $C/(v_n^2 d)$ and $C\xi_n/v_n^2$, respectively; both vanish. Each true class contains a row outside both sets. The K true profiles are separated in ℓ_1 norm by cd, so their nearby fitted profiles must be different. There are exactly K fitted profiles, giving the asserted common matching.

Fix a sufficiently small constant t_0. Except for Cn/d rows, the row's ℓ_1 distance to its true profile is at most $t_0 d$, by the Frobenius bound. For every such row in class a, matched-center consistency and box Lipschitz bounds imply

$$D(q_i \| \mu_b) \geq c_1 d \quad (b \neq a), \qquad D(q_i \| \mu_a) \leq c_1 d/4.$$

Each class has at least $\pi_0 n/2$ such rows for large n. Since each summand of $\mathcal{F}$ is nonnegative and on these rows is at least $-\log(\pi_a + e^{-c_1 d})$, we obtain

$$\pi_a + e^{-c_1 d} \geq e^{-2\xi_n d/\pi_0}, \qquad \pi_a \geq \tfrac{1}{2} e^{-2\xi_n d/\pi_0}$$

eventually. The latter holds whether or not $\xi_n d$ diverges, since $\xi_n \to 0$. Thus $-\log \pi_a \leq 2\xi_n d/\pi_0 + \log 2 = o(d)$. Apply Lemma 4 to the fit itself. □

Proposition 6 (The current replacement rule supplies the objective certificate)**.** *Under* (1)*, the existing deterministic growing-and-replacement algorithm with* $T = \lceil \log n \rceil$ *replacement rounds has*

$$\frac{\mathcal{F}_T}{n^2 p} \leq C(1 - 1/K)^T + \frac{C}{\sqrt{np}} + \frac{K \log K}{np} = o(1)$$

on the exponentially reliable graph event. Early stopping after a full round without improvement also certifies $\mathcal{F} = o(n^2 p)$*.*

Proof. Write $d = np$. The Frobenius bound gives $\sum_i \|q_i - P_{z_i,*}\|_1 \leq n\|q - M_0\|_{\mathrm{F}} \leq Cn\sqrt{d}$. Define

$$G(\mathcal{C}) = \sum_i \min_k D(q_i \| \mu_k), \qquad G_*(\mathcal{C}) = \sum_a n_a \min_k D(P_{a*} \| \mu_k).$$

The divergence is uniformly Lipschitz in ℓ_1 on the common box, so $|G - G_*| \leq Cn\sqrt{d}$ simultaneously over every admissible center list. Each true class supplies an observed candidate within $C\sqrt{d}$ in ℓ_1 of its true profile.

For any K fitted centers, choose a minimizing center for each true class. A class contributing at least G_*/K exists. If the chosen-center map is not onto, delete an unused center; otherwise it is a bijection, and delete the center chosen by this class. Inserting its true profile removes at least that class's contribution and harms none of the other classes. Thus some ideal replacement decreases population hard loss by at least G_*/K. Replacing the ideal profile by its available observed candidate and applying the uniform Lipschitz bounds gives an observed trial with hard loss at most $(1 - 1/K)G + Cn\sqrt{d}$.

For each deletion, the implemented maximum positive all-node score gain minimizes exactly the resulting hard loss. All deletions are tested. Uniformly initializing that trial's weights changes

its soft loss from hard loss by at most $n \log K$. Exact E/M fitting decreases soft loss; selecting the best trial together with the unchanged fit gives

$$\mathcal{F}_{t+1} \le (1 - 1/K)\mathcal{F}_t + Cn\sqrt{d} + n \log K, \qquad \mathcal{F}_{t+1} \le \mathcal{F}_t.$$

Initially $\mathcal{F}_0 \le Cnd$ for arbitrary simplex weights, since all profile divergences are $O(d)$ on the common box. Iterate the recurrence. If a complete round does not improve the fit, the same inequality gives $\mathcal{F}_t \le K(Cn\sqrt{d} + n \log K)$ directly. □

Corollary 7 (The existing certified algorithm attains the Zhou–Li scale)**.** *Under* (1)*, the existing growing-and-replacement algorithm followed by its mandatory final E/M update, with no additional algorithmic change, satisfies*

$$\mathbb{E}\,\mathrm{Mis}(\widehat{z}, z) \le C\frac{e^{-I_n}}{\sqrt{np}}.$$

Proof. Proposition 6 and Lemma 5 show that the mandatory final E-step already supplies responsibilities with error $C/(np)$. The subsequent mandatory M-step therefore meets Lemma 8. The refined risk analysis below applies directly, without first taking the extra bootstrap update needed for a merely weak responsibility initializer. □

4 Fitting costs only a constant-scale global fluctuation

Define the perfect-label fitted mean $\bar{\mu}_a = n_a^{-1} \sum_{l:z_l=a} q_l$. Let

$$G = \max_{a,c} \left| \frac{\mathbf{1}_{C_a}^\top (A - M)\mathbf{1}_{C_c}}{\sqrt{n_a n_c}} \right|.$$

For each fixed t, $\mathbb{E}e^{tG} \le C_t$. Indeed, these are finitely many independent-edge sums with coefficient $O(n^{-1})$ and variance $O(p)$; the usual bounded-variable moment generating function estimate applies to both signs.

Lemma 8 (Fitted-score comparison)**.** *On the full-graph good event, every M-step with aligned responsibility error at most $C/(np)$ satisfies, uniformly in i, a, b,*

$$\left| \ell_i(\mu_a) - \ell_i(\mu_b) + \log\frac{\pi_a}{\pi_b} - \left\{ \sum_j q_{ij} w_{z_j}^{ab} + \sum_j b_{z_j}^{ab} \right\} \right| \le C(1 + G). \tag{3}$$

Proof. The mean-fitting identity and the bound $\|q_l\|_1 \le Cnp$ show $\|\mu_a - \bar{\mu}_a\|_1 \le C$. Score derivatives in a parameter coordinate are $(q_{ij} - \mu_j)/[\mu_j(1 - \mu_j)]$ and are bounded on the common box, so this replacement costs $O(1)$. Updated log-weight ratios are bounded as well.

Write $\delta_a = \bar{\mu}_a - P_{a*}$. Cauchy–Schwarz and the Frobenius bound give $\|\delta_a\|_2^2 \le Cp$. Also

$$\|q_i(I - \Pi)\|_2 \le C\sqrt{p}. \tag{4}$$

For F, this follows from row incoherence, signal size $O(np)$, and projector error $O((np)^{-1/2})$. For q it follows because a common scalar 1-Lipschitz map contracts within-block variance: use the pairwise formula for variance in each block.

We next bound each block sum of δ_a. With $R = A - F$, one has $R = (I - UU^\top)R(I - UU^\top)$ and $\|R\|_{\mathrm{op}} \le C\sqrt{np}$. Consequently

$$\|\Pi R \Pi\|_{\mathrm{op}} \le C\sqrt{np}\,\|\Pi - UU^\top\|_{\mathrm{op}}^2 \le C/\sqrt{np}.$$

Thus block-averaging F instead of A changes the relevant block sum by at most a constant. Raw block averaging contributes at most $C(G+1)$, including the absent diagonal. Finally, if a true probability u is a distance at least $c'p$ from both clipping endpoints, then

$$|\operatorname{clip}(x,\epsilon,1-\epsilon)-x| \le \frac{(x-u)^2}{c'p}.$$

Summing this inequality and averaging over class a gives $\|\bar{\mu}_a - n_a^{-1}\mathbf{1}_{C_a}^\top F\|_1 \le C$. We have proved

$$\left|\sum_{j\in C_c}\delta_{aj}\right| \le C(1+G). \tag{5}$$

Taylor-expand ℓ_i in its parameter around P_{a*}. Its second derivatives are bounded by C/p, so the quadratic remainder is at most $C\|\delta_a\|_2^2/p \le C$. Split the linear term into the true block span and its orthogonal complement. The block part is bounded by (5); the derivative coefficient is block constant and bounded. The orthogonal part is bounded by $C\|q_i(I-\Pi)\|_2\|\delta_a\|_2/p \le C$ using (4). Apply this to both classes. □

5 An exact arrowhead reduction of empirical spectral block sums

Fix vertex i and condition on $B = A_{-i,-i}$. Its incident-edge vector $x = A_{-i,i}$ remains independent Bernoulli with block-constant probabilities. Let V, D be the retained signal eigenpairs of B, write $N = B - VDV^\top$, and let $\Pi_B = VV^\top$. Let Π_0 be the true block projector on the remaining vertices. On the vertex-deleted good event,

$$\|N\|_{\mathrm{op}} \le C\sqrt{np}, \quad \|V\|_{2\to\infty} \le C/\sqrt{n}, \quad \|\Pi_B - \Pi_0\|_{\mathrm{op}} \le C/\sqrt{np}.$$

All assertions below about x are deterministic on $\|x\|_1 \le D_0 np$, with fixed sufficiently large D_0.

Lemma 9 (Arrowhead linearization with bounded remainder)**.** *For every fixed vector $w \in \mathbb{R}^n$ with $\|w\|_\infty \le W$,*

$$\left|F_i w - x^\top \Pi_B w_{-i}\right| \le C_W. \tag{6}$$

The constant is uniform over all incident-edge vectors satisfying the degree bound.

Proof. Permute i to the first coordinate and write $A = \left(\begin{smallmatrix} 0 & x^\top \\ x & B \end{smallmatrix}\right)$. The difference from $\operatorname{diag}(0,B)$ has norm $\|x\|_2 \le \sqrt{D_0 np}$. Weyl's inequality consequently gives exactly K eigenvalues of magnitude $\asymp np$, with all remaining eigenvalues $O(\sqrt{np})$. This assertion uses good B and the degree bound alone. The retained eigenvectors can be written $u_r = (t_r, Vy_r + v_r)$ with $v_r \perp V$. The exact eigenvector equations give

$$v_r = t_r(\lambda_r - N)^{-1}x_Q, \qquad (\lambda_r - h_r)t_r = x_V^\top y_r, \quad h_r = x_Q^\top(\lambda_r - N)^{-1}x_Q,$$

where $x_V = V^\top x$, $x_Q = (I-\Pi_B)x$, and $|h_r| \le C$.

The padded basis $V_0 = (0, V^\top)^\top$ has spectral residual $AV_0 - V_0 D = (x_V^\top, 0)^\top$, of norm at most $Cnp/\sqrt{n}$. For a full-graph bulk eigenvector h and column $V_{0,r}$,

$$(\lambda_h - D_r)\langle h, V_{0,r}\rangle = \langle h, (AV_0 - V_0 D)_r\rangle.$$

The denominator has magnitude at least cnp, for either signal sign. Sum the squared identity over bulk eigenvectors and the fixed K columns to obtain $\|(I - UU^\top)V_0\|_{\mathrm{F}} \le C/\sqrt{n}$. Equality of the two subspace dimensions gives $\|UU^\top - V_0V_0^\top\|_{\mathrm{op}} \le C/\sqrt{n}$. In particular

$$\sum_r t_r^2 \le C/n, \qquad \left\|\sum_r y_r y_r^\top - I_K\right\|_{\mathrm{op}} \le C/n.$$

The latter uses squared principal angles, not the unsquared projector bound.

Put $w_V = V^\top w_{-i}$ and $w_Q = (I - \Pi_B)w_{-i}$. Expanding the retained spectral sum gives

$$\begin{aligned} F_i w = x_V^\top \left(\sum_r y_r y_r^\top\right) w_V + \sum_r h_r t_r y_r^\top w_V \\ + \sum_r \lambda_r t_r^2 \{w_i + x_Q^\top (\lambda_r - N)^{-1} w_Q\}. \end{aligned}$$

The errors relative to $x_V^\top w_V$ are bounded respectively by

$$C(np/\sqrt{n})n^{-1}\sqrt{n} = Cp, \qquad Cn^{-1/2}\sqrt{n} = C, \qquad C(np/n)\{1 + \sqrt{np}\sqrt{n}/(np)\} \le C.$$

This proves (6) without assuming independence between the full-graph eigenvectors and x. □

Lemma 10 (Projected block weights)**.** *For any true block-constant vector v on the remaining vertices with bounded entries, let $\beta = (\Pi_B - I)v$. Then*

$$\|\beta\|_\infty \le C, \quad \|\beta\|_2^2 \le C/p, \quad \left|\sum_{j \in C_c \setminus \{i\}} \beta_j\right| \le C/p.$$

Proof. The first follows from V's row bound and $\|v\|_2 = O(\sqrt{n})$, and the second from the projector bound. For the third, since $v \in \mathrm{col}(\Pi_0)$,

$$\Pi_0\beta = -\Pi_0(I - \Pi_B)\Pi_0 v, \qquad \|\Pi_0(I - \Pi_B)\Pi_0\|_{\mathrm{op}} \le \|\Pi_0 - \Pi_B\|_{\mathrm{op}}^2 \le C/(np).$$

Take its inner product with a block indicator. □

6 Clipping and the refined conditional tail

Choose a sufficiently small fixed $h > 0$ such that $2c_0 + h < \min(\kappa, \eta)/2$, and put

$$l = (2c_0 + h)p, \qquad u = 1 - (2c_0 + h)p.$$

For each full block C_c set $T_c = F_i \mathbf{1}_{C_c}$ and $Y_c = x^\top \Pi_B \mathbf{1}_{C_c \setminus \{i\}}$.

Lemma 11 (Deterministic buffered clipping envelopes)**.** *On the full-graph and vertex-deleted good events and the degree event,*

$$\begin{aligned} \sum_{j \in C_c} q_{ij} &\ge \min(Y_c, n_c u) - C, \\ \sum_{j \in C_c} q_{ij} &\le \max(Y_c, n_c l) + C. \end{aligned}$$

Proof. Write $F_{ij} = T_c/n_c + v_j$ in block c. The v_j sum to zero and (4) for F gives $\sum v_j^2 \le Cp$. If the block mean is at least hp below an upper clipping endpoint, its upper clipping loss is at most $\sum(v_j - hp)_+ \le \sum v_j^2/(4hp) \le C$. If it is above that buffered endpoint, monotonicity gives the same lower envelope with the block mean replaced by the buffered endpoint. The lower endpoint is treated symmetrically. Use $\epsilon \le 2c_0 p$ and then $|T_c - Y_c| \le C$ from Lemma 9. □

For a fixed pair a, b, abbreviate $w_c = w_c^{ab}$, $b_c = b_c^{ab}$, and define

$$H(x) = \sum_{w_c \ge 0} w_c \min(Y_c, n_c u) + \sum_{w_c < 0} w_c \max(Y_c, n_c l) + \sum_c n_c b_c.$$

The true-profile score based on q is at least $H(x) - C$. For each clipping pattern $S \subseteq [K]$, where S denotes the unclipped coordinates, its branch is

$$H_S(x) = x^\top \Pi_B v_S + \sum_{c \notin S} w_c k_c + \sum_c n_c b_c, \quad (v_S)_j = w_{z_j} \mathbf{1}\{z_j \in S\},$$

with $k_c = n_c u$ for $w_c \ge 0$ and $k_c = n_c l$ for $w_c < 0$. One has $H(x) = H_S(x)$ on the appropriate branch region; dropping that region gives an upper bound on each branch's error probability.

Lemma 12 (Conditional cumulant and concentration bounds)**.** *Let $\alpha = \alpha^*_{i,ab}$. Uniformly over good B and every branch,*

$$\mathbb{E}_a[e^{-\alpha H_S(x)} \mid B] \le C e^{-I_{i,ab}}. \tag{7}$$

If S contains a block with $|w_c| \ge \tau$, for a fixed $\tau > 0$, then, for every real t,

$$\mathbb{P}_a\{H_S(x) \le t \mid B\} \le \frac{C_\tau e^{-I_{i,ab} + \alpha t}}{\sqrt{np}}. \tag{8}$$

Proof. First replace $\Pi_B v_S$ by v_S. Write $m_c = n_c - \mathbf{1}\{z_i = c\}$. The ideal counts in different blocks are independent Binomial(m_c, P_{ac}). For sufficiently large n, the lower cutoff satisfies $n_c l \le m_c P_{ac}$, including the deleted vertex's class, since $l < \kappa p/2$ and $n_c \ge \pi_0 n$. The upper cutoff is above its mean as well. Thus each omitted coordinate satisfies $w_c k_c \ge w_c \mathbb{E}_a \sum_{j \in C_c \setminus \{i\}} x_j$. Jensen's inequality bounds its deterministic exponential factor by that block's full exponential moment. The ideal branch moment is at most $e^{-I_{i,ab} + O(p)}$: the branch and oracle intercepts differ by exactly $\sum_c (n_c - m_c) b_c = b_{z_i} = O(p)$.

For the empirical projection put $\beta = (\Pi_B - I) v_S$. Taylor-expand the log moment as a function of its coefficient vector. At the ideal block-constant coefficients its first derivative is a bounded multiple of a block-constant tilted probability, of order p. The block-sum bound in Lemma 10 makes the linear change $O(1)$. Along the segment to the perturbed coefficients every second derivative is bounded by Cp, since all coefficients are bounded. The quadratic change is at most $Cp\|\beta\|_2^2 \le C$. This proves (7).

Under exponential tilting by $e^{-\alpha H_S}$ the coordinates of x are still independent Bernoulli, with probabilities comparable to p and complements bounded below. If a block in S has $|w_c| \ge \tau$, then $\|v_S\|_2^2 \ge c_\tau n$, while $\|\beta\|_2^2 \le C/p = o(n)$. The variance of $x^\top \Pi_B v_S$ under this tilted law is therefore between $c_\tau np$ and Cnp. Bounded coefficients give a total third absolute centered moment at most a constant times this variance. Berry–Esseen consequently bounds the probability of every interval of length one by $C_\tau/\sqrt{np}$.

Divide $(-\infty, t]$ into intervals $(t - r - 1, t - r]$. Changing measure back gives

$$\mathbb{P}_a(H_S \le t \mid B) \le \mathbb{E}_a(e^{-\alpha H_S} \mid B) \frac{C_\tau e^{\alpha t}}{\sqrt{np}} \sum_{r \ge 0} e^{-\alpha r}.$$

The maximizing α is uniformly bounded away from zero. This proves (8); no nonlattice assumption or exact boundary density is used. □

Some branches can have no strongly weighted unclipped coordinate. They cannot cause an error on the graph event. In fact the oracle mean LLR is at least $c_1 np$. Omitted coordinates replace their means in the score-increasing direction. On $\|x\|_1 \le D_0 np$, $|Y_c| \le C D_0 np$. Therefore, if every unclipped weight has magnitude less than τ,

$$H_S(x) \ge c_1 np - K\tau(CD_0 + 1)np - O(1) \ge c_1 np/2$$

for a sufficiently small fixed τ. Thus all potentially erroneous branches have the concentration required by Lemma 12.

7 Completing the refined risk bound

Let G_i denote the block fluctuation G computed on the vertex-deleted graph B, with the corresponding block sizes. On the good graph and degree event,

$$G \le C(1 + G_i), \qquad G_i \le C\sqrt{np}, \qquad \mathbb{E}e^{tG_i} \le C_t \quad (t \text{ fixed}).$$

The first bound holds because adding back one row changes a normalized block sum by at most $C \deg(i)/n + Cp \le C$.

After a M-step from the $C/(np)$ responsibility radius (either M_2 in Theorem 3, or the existing final M-step in Corollary 7), Lemma 8 shows that assigning vertex i from a to b implies

$$H(x) \le C(1 + G_i).$$

First use the full-graph good event only to establish this deterministic containment of the error event. Then remove its indicator from the upper bound; it is not assumed independent of x. Conditional on good B, branches with only weak unclipped weights are impossible on the degree event for large n, since $G_i = O(\sqrt{np})$. For every other branch apply (8), with threshold $C(1 + G_i)$. Integrating over B and using its bounded exponential moment yields

$$\mathbb{P}_a\{\text{aligned error at } i, \text{ good events}\} \le C\frac{e^{-I_{i,ab}}}{\sqrt{np}}.$$

The union over $b \ne a$ and the at most 2^K branches costs a fixed constant. The graph, vertex-deleted graph, and degree failures are $C_H e^{-Hnp}$. Since $I_n \le C_I np$, choose fixed $H > C_I + 1$; their total contribution is negligible relative to $e^{-I_n}/\sqrt{np}$. Average over vertices and include the initialization failure. This proves (2) subject to the stated spectral input. The invariant bootstrap radius proves the additional-update claim.

8 Scope and independent checks

1. This is a constant-factor rate, not a claim that the risk ratio tends to one. Bounded spectral and fitting corrections need not vanish, and Bernoulli ties can affect the leading constant.

2. A merely weak responsibility initializer needs the local two-M-step refinement. The certified algorithm has more information: its low objective loss makes its own final E-step already accurate to $C/(np)$. Consequently its existing mandatory M-step suffices; no extra iteration or raw-adjacency access is required for Corollary 7.

3. Proposition 2 spells out the exponential-confidence specialization of standard matrix concentration and entrywise eigenspace results. Proposition 6 re-establishes the current general-K replacement certificate using the simpler Frobenius bound. The external spectral theorems themselves are cited, not reproved here.

4. The new arrowhead argument, buffered clipping argument, uniform branch cumulant expansion, and fitted-score comparison have undergone independent internal mathematical checks. A separate 144-graph conditional tilted-row calculation checks the arrowhead identity and bounded-remainder scaling over multiple sizes and signal signs. It is a diagnostic, not a proof of the exponentially small risk theorem.

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